F18CD Real Analysis

Week 1

What is Calculus?

Calculus is the mathematical study of continuous change.

Calculus has two major branches, differential calculus and integral calculus; the former concerns instantaneous rates of change, and the slopes of curves, while the latter concerns accumulation of quantities, and areas under or between curves.

What is Analysis?

Analysis is the branch of mathematics dealing with continuous functions, limits, and related theories, such as differentiation, integration, measure, infinite sequence, series, and analytic functions.

These theories are usually studied in the context of real and complex numbers and functions.

What is the relation between Calculus and Real Analysis?

Analysis evolved from calculus, which involves the elementary concepts and techniques of analysis. Analysis may be distinguished from geometry; however, it can be applied to any space of mathematical objects that has a definition of nearness (a topological space) or specific distances between objects (a metric space).

Why we need to learn Analysis?

Lecture notes Chapter 1 (page 1 – 4) provide some motivations and examples.

Problem 1. Let investigate the infinite sum (alternating harmonic series)

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots.$$

We obtain:

$$\sum_{n=1}^{100} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{98} + \frac{1}{99} - \frac{1}{100} \approx 0.688 \text{ (summing the first 100 terms)},$$

$$\sum_{n=1}^{500} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{498} + \frac{1}{499} - \frac{1}{500} \approx 0.692 \text{ (summing the first 500 terms)},$$

In fact, the answer is ln 2 if we continue the summing to infinity.

However, we *may* arrive at another answer:

Rearrange the infinite series as

$$1 + \tfrac{1}{3} - \tfrac{1}{2} + \tfrac{1}{5} + \tfrac{1}{7} - \tfrac{1}{4} + \ldots$$

Summing the first 100 terms gives ≈ 1.032.

Summing the first 500 terms gives ≈ 1.038.

So which answer is correct? Anything went wrong?

Here is where Analysis came in to rescue! We will study rigorous definitions and proofs to provide solid foundations for mathematics.

Number systems:

Here are the standard abbreviations for the several kind of numbers.

$\mathbb{N} := \{1, 2, 3, \ldots\}$	the set of *natural numbers*
$\mathbb{Z} := \{0, -1, 1, -2, 2, -3, 3, \ldots\}$	the set of *integers*
$\mathbb{Z}^+ := \mathbb{N} = \{1, 2, 3, \ldots\}$	the set of *positive integers*
$\mathbb{Z}^- := -\mathbb{N} = \{-1, -2, -3, \ldots\}$	the set of *negative integers*
$\mathbb{Q} := \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$	the set of *rational numbers*

What is *dense*?

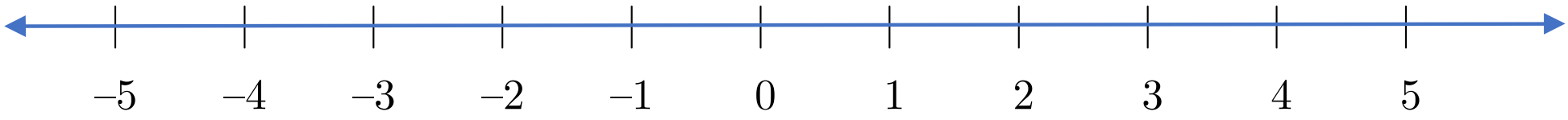
$\mathbb{Q}$ has a very special property that $\mathbb{N}$ and $\mathbb{Z}$ don't have:

> Between any $x, y \in \mathbb{Q}$, there exists $z \in \mathbb{Q}$ such that $x < z < y$.

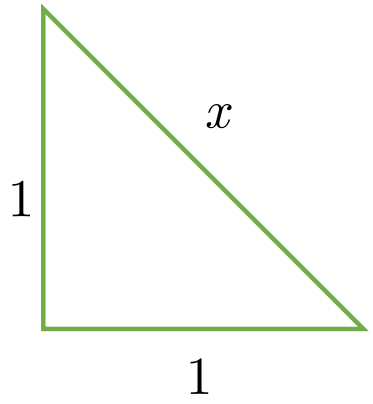
It is easy to find z, take $z = \frac{x+y}{2}$ (we call this as bisection).

The mathematical term for this is to say that the rational numbers are *dense* – means we can *always* find a rational number between any two rational numbers, and we can continue this process forever!

Given this property, it seems quite reasonable to represent rational numbers by a straight line extending infinitely from left to right, as seen below:



If this were the case, then it would suggest that the rational numbers are *sufficient* for measuring all quantities, such as geometric lengths. *But* what happens when we try to find the length x? In other words, is x a rational number?



Theorem 2.4. $\sqrt{2}$ is not a rational number.

Hint: n is even if and only if n^2 is even (whenever n^2 is a perfect square).

Proof (by contradiction). Suppose $\sqrt{2}$ is a rational number. Then

$$\sqrt{2} = \frac{m}{n}, \text{ where } m \text{ and } n \text{ are positive integers.}$$

Without losing its generality, assume that $\gcd(m, n) = 1$ (the fraction $\frac{m}{n}$ is in the simplest form, or irreducible).

So, we have $2n^2 = m^2$. Note that

$$2 \mid m^2 \Rightarrow 2 \mid m \text{ (from hint: } n^2 \text{ is even iff } n \text{ is even).}$$

So there exists integer k such that $m = 2k$. Substitute $m = 2k$ into $2n^2 = m^2$, we obtain $n^2 = 2k^2$ which means $2 \mid n$ by the similar argument as above.

We now arrive the conclusion that both m and n are even, which mean the fraction is not in the simplest fraction form, i.e., $\gcd(m, n)$ not equal to 1; a contradiction.

New kinds of number beyond rational numbers?

We must now face the hard truth that the rational numbers are not sufficient to represent the measurement of all lengths.

The story of Hippasus: Hippasus - Wikipedia

Extension of $\mathbb{Q}$

So, we must extend/complete $\mathbb{Q}$ so that we are able to assign numerical values to any measurement.

We start to identify all the properties of the rational numbers, so that the new number system, should it even exist, must minimally satisfy all the same properties of the rational numbers (or plus some more). This leads to the following definition.

Binary operation

Definition.

Let S be a nonempty set. A *binary operation* on S is a function from $S \times S$ to S.

If $*$ is a binary operation on S, we write $x * y$, where $x, y \in S$ and we say that S is closed under $*$.

Example:

(a) $\mathbb{Z}$ is closed under $+$ since if we add two integers we obtain an integer. So we say that $+$ is a binary operation on $\mathbb{Z}$.

(b) $\mathbb{N}$ is not closed under $-$ (subtraction) since $2 - 3 = -1 \notin \mathbb{N}$. So $-$ is not a binary operation on $\mathbb{N}$.

Mathematical system

Definition.

Let S be a nonempty set. If $*$ is a binary operation on S. Then we say that $\left(S,*\right)$ is a *mathematical system*.

Example: $\left(\mathbb{Z},+\right),\left(\mathbb{Z},\times\right)$ are mathematical system.

Ordered field

Definition 2.5

An *ordered field* $\mathbb{F}$ is a non-empty set with two binary operations, addition $(+)$ and multiplication $(\cdot)$, as well as two relations $(<\ )$ and $(=)$, such that for all $a, b, c \in \mathbb{F}$, there exist $0, 1 \in \mathbb{F}$, that the following are satisfied:

(i)	(Commutative Law)	$a+b=b+a$ and $a\cdot b=b\cdot a$.
(ii)	(Distributive Law)	$a\cdot(b+c)=a\cdot b+a\cdot c$.
(iii)	(Associative Law)	$a+(b+c)=(a+b)+c$ and $a\cdot(b\cdot c)=(a\cdot b)\cdot c$.
(iv)	(Identity Law)	$a+0=a$ and $a\cdot 1=a \quad \forall a\in\mathbb{F}$.
(v)	(Inverse Law)	$\forall a\in\mathbb{F}$, $\exists -a\in\mathbb{F}$ such that $a+(-a)=0$. If $a\neq 0$, there $\exists a^{-1}\in\mathbb{F}$ such that $a\cdot a^{-1}=1$.
(vi)	(Order Axioms)	(a) Exactly one is true: Either $a<b, b<a,$ or $a=b$. (b) $a<b\Rightarrow a+c<b+c \quad \forall c\in\mathbb{F}$. (c) $a,b>0\Rightarrow a\cdot b>0$. (d) $b<a \;\&\; c<b\Rightarrow c<a$.
$\mathbb{F}$ is called a *field* if it satisfies (i) to (v) (exclude (vi)).		

Examples

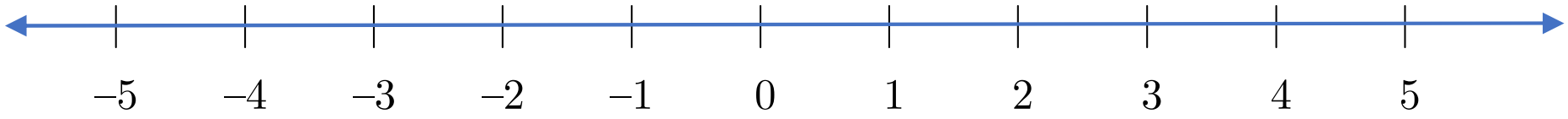
$\left(\mathbb{Z}, +, \times\right)$ is not a field.

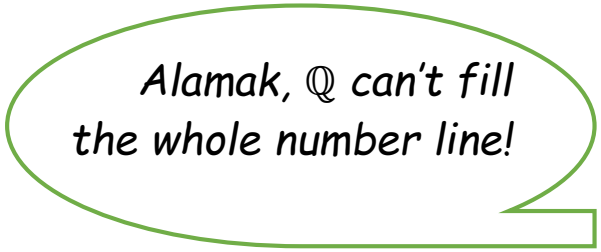
$\left(\mathbb{Q}, +, \times\right)$ is a field.

$\left(\mathbb{R}, +, \times\right)$ and $\left(\mathbb{C}, +, \times\right)$ are fields.

Filling in the gaps

We have now seen that ordered fields are insufficient for describing continuous measurements since ordered fields like $\mathbb{Q}$ contains infinitely many holes.





In order to find a way to counter this problem, we need some new "weapons".

Let $\mathbb{F}$ be an ordered field, and let S be a non-empty subset of $\mathbb{F}$.

Definition 2.6: Bounded Set

Then S is said to be *bounded above* if $\exists \alpha \in \mathbb{F}$ such that $\alpha \geq x \ \forall x \in S$.

(α is called an upper bound for S).

Then S is said to be *bounded below* if $\exists \beta \in \mathbb{F}$ such that $\beta \leq x \ \forall x \in S$.

(β is called a lower bound for S).

S is bounded above:	$\exists \alpha \in \mathbb{F}$ such that $\alpha \geq x \ \forall x \in S$.	α is an upper bound.
S is bounded below:	$\exists \beta \in \mathbb{F}$ such that $\beta \leq x \ \forall x \in S$.	β is a lower bound.

Remark	Does it always exist?	Unique (if it exist)?	An element of the set?
Max/min	No	Yes	Yes
Upper/lower bound	No	No	Not necessary
Sup/inf	Yes if full the theorem	Yes	Not necessary

Let $\mathbb{F}$ be an ordered field, and let S be a non-empty subset of $\mathbb{F}$.

Definition 2.7(a)

If S is bounded above. Then the supremum of S (is an element of $\mathbb{F}$), should it exist, is the least upper bound of S and is denoted by $\sup S$.

Remarks:

1. If $\sup S$ exists, then
   (a) $\sup S$ is an upper bound, i.e., $\sup S \geq x \ \forall x \in S$, *and*
   (b) $\sup S$ is the smallest upper bound, i.e., U is an upper bound $\Rightarrow \sup S \leq U$.
2. If S has no upper bound then for sure $\sup S$ does not exist.
3. $\sup S \notin S$ (may not be in S).
4. $\sup S$ is unique if it exists.

The important fact is, when will sup S be an element of $\mathbb{F}$?

Let $\mathbb{F}$ be an ordered field, and let S be a non-empty subset of $\mathbb{F}$.

Definition 2.7(b)

If S is bounded below. Then the infimum of S (is an element of $\mathbb{F}$), should it exist, is the greatest lower bound of S and is denoted by $\inf S$.

Remarks:

1. If $\inf S$ exists, then
   (a) $\inf S$ is a lower bound, i.e., $\inf S \leq x \ \forall x \in S$, *and*
   (b) $\inf S$ is the largest lower bound, i.e., L is a lower bound $\Rightarrow \inf S \geq L$.
2. If S has no lower bound then for sure $\inf S$ does not exist.
3. $\inf S$ may not be an element of S.
4. $\inf S$ is unique if it exists.

The important fact is, when will $\inf S$ be an element of $\mathbb{F}$?

Remark	Does it always exist?	Unique (if it exist)?	An element of the set?
Max/min	No	Yes	Yes
Upper/lower bound	No	No	Not necessary
Sup/inf	No	Yes	Not necessary

Example: Let $S = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$. Then $\sup S = 1$ and $\inf S = 0$.

Note that $1 \in S$ but $0 \notin S$.

Question 1

1. What is the max/min of a set S? S is a subset of an ordered field.

The largest/smallest element in S; if it exists.

Example: $A = \{n : n \in \mathbb{N}\}$ then $\min A = 1$ but $\max A$ does not exist.

Question 2

2. What is an upper/lower bound of a set S? S is a subset of an ordered field.

A number (not unique) that is bigger/smaller or equal than every element in S, if it exists.

Example: $A = \{n : n \in \mathbb{N}\}$ then any ≤ 1 are lower bound of A.

Remark: A is not bounded from above so there is no upper bound for A.

Question 3

3. What is the sup/inf of a set S? S is a subset of an ordered field $\mathbb{F}$.

Sup (is an element of $\mathbb{F}$), if it exists, is the least upper bounds.

Inf (is an element of $\mathbb{F}$), if it exists, is the largest lower bounds.

Sup/inf may not always be an element of the set.

Sup/inf may not even exist in $\mathbb{F}$, but if is, then it must be unique.

Example: $A = \{n : n \in \mathbb{N}\}$ then 1 is the inf A.

Remark: sup A does not exist because there is no upper bound for the set.

Question 4

4. Does sup/inf always exists?

Not all the time, but under The Completeness Axiom, then the sup/inf exists.

In other words, if $\mathbb{F}$ is a complete ordered field, then sup/inf exists.

Definition 2.8: Complete ordered field

An ordered field $\mathbb{F}$ is said to be *complete* if for every subset $S \subset \mathbb{F}$ that is bounded above, there exists a supremum in $\mathbb{F}$.

Remark: Equally, completeness could be defined as every subset $S \subset \mathbb{F}$ that is bounded below has an infimum in $\mathbb{F}$.

Theorem 2.9: Real Numbers

There exists a *unique* complete ordered field.

Remark: We then give this a special name – the real field and award a special symbol for it: $\mathbb{R}$ as the set of real numbers.

Remarks:

1. Unfortunately (or fortunately), we will not be proving this theorem. *This is studied in abstract algebra.*

2. The word unique here would mean that every complete ordered field we can find is the same, or more accurately *isomorphic* to $\mathbb{R}$. *This is studied in abstract algebra.*

3. Mathematically, the real numbers are said to satisfy a universal property. *This is studied in abstract algebra.*

We now may define the irrational numbers as well:

$$\mathbb{I} = \mathbb{Q}' := \mathbb{R} \setminus \mathbb{Q} \qquad \text{the set of } \textit{irrational numbers}$$

$\sqrt{2}, \pi, e, \cos\frac{\pi}{15}$ are examples of irrational numbers.

In fact, numbers like π, e are called *transcendental* number, or *non-algebraic* numbers.

Corollary 2.10 Real numbers

An real numbers $\mathbb{R}$ is a non-empty set with two binary operations, addition $(+)$ and multiplication $(\cdot)$, as well as two relations $(<)$ and $(=)$, such that for all $a, b, c \in \mathbb{R}$, there exist $0, 1 \in \mathbb{R}$, that the following are satisfied:

(i)	(Commutative Law)	$a + b = b + a$ and $a \cdot b = b \cdot a$.
(ii)	(Distributive Law)	$a \cdot (b + c) = a \cdot b + a \cdot c$.
(iii)	(Associative Law)	$a + (b + c) = (a + b) + c$ and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
(iv)	(Identity Law)	$a + 0 = a$ and $a \cdot 1 = a \quad \forall a \in \mathbb{R}$.
(v)	(Inverse Law)	$\forall a \in \mathbb{R}$, $\exists -a \in \mathbb{R}$ such that $a + (-a) = 0$. If $a \neq 0$, there $\exists a^{-1} \in \mathbb{R}$ such that $a \cdot a^{-1} = 1$.

(vi)	(Order Axioms)	(a) Exactly one is true: Either $a < b, b < a,$ or $a = b$. (b) $a < b \Rightarrow a + c < b + c \quad \forall c \in \mathbb{R}$. (c) $a, b > 0 \Rightarrow a \cdot b > 0$. (d) $b < a \;\&\; c < b \Rightarrow c < a$.
(vii)	(Completeness Axiom)	$\forall S \subset \mathbb{R}$ that is bounded above, there exists $\sup S \in \mathbb{R}$.

Final remarks

- $\mathbb{R}$ satisfies the completeness axiom which makes it distinct from $\mathbb{Q}$.
- We will see the direct consequences of the completeness axiom.
- We emphasise that every result proven in this course are from axioms here.
- This is fundamentally how modern mathematics is done: that is, the enormity of theorems and lemmas proven in any mathematical discipline come from the fewest number of assumptions, or axioms, that can be made.

Properties of the Real Numbers

Definition 3.1 Modulus/Absolute value

The absolute value function $|\cdot|: \mathbb{R} \to \mathbb{R}$ is defined as

$$\left|x\right| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}.$$

Example 3.2.

(i) $|7| = 7$ (ii) $|{-5}| = 5$ (iii) $|0| = 0$.

Absolute value involving inequalities

Case 1: $\left|x-a\right| \le b \Leftrightarrow -b \le x-a \le b \Leftrightarrow a-b \le x \le a+b$

Case 2: $\left|x-a\right| \ge b \Leftrightarrow x-a \le -b$ **or** $x-a \ge b \Leftrightarrow x \le a-b$ **or** $x \ge a+b$

Remark: The result is true when $\le$ is replaced by $<$, and $\ge$ is replaced by $>$.

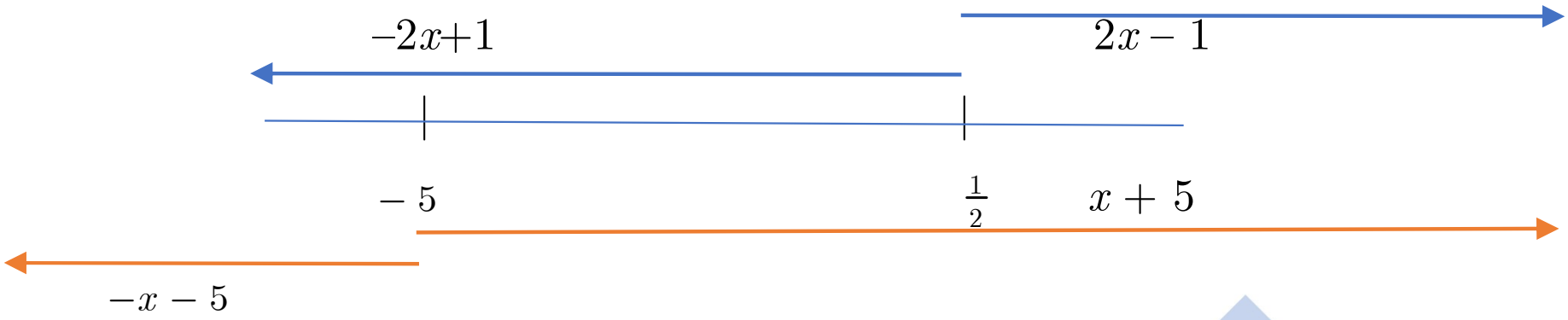
Example.

$$\begin{aligned}\left|x-3\right| \le 0.2 &\Rightarrow -0.2 \le x-3 \le 0.2 \\ &\Rightarrow 3-0.2 \le x \le 3+0.2 \\ &\Rightarrow 2.8 \le x \le 3.2\end{aligned}$$

Self-study

Solve $|2x-1|-|x+5|<10$.

Since $|2x-1|=\begin{cases}2x-1, & x\geq\frac{1}{2}\\ -(2x-1), & x<\frac{1}{2}\end{cases}$, and $|x+5|=\begin{cases}x+5, & x\geq -5\\ -(x+5), & x<-5\end{cases}$.



$$|2x-1|-|x+5| = \begin{cases} 2x-1-(x+5), & x \geq \frac{1}{2} \\ -(2x-1)-(x+5), & -5 \leq x < \frac{1}{2} \\ -(2x-1)+(x+5), & x < -5 \end{cases} = \begin{cases} x-6, & x \geq \frac{1}{2} \\ -3x-4, & -5 \leq x < \frac{1}{2} \\ -x+6, & x < -5 \end{cases}$$

$$|2x-1|-|x+5|<10$$

$$\Rightarrow\begin{cases} x-6<10, & x\geq\frac{1}{2} \\ -3x-4<10, & -5\leq x<\frac{1}{2} \\ -x+6<10, & x<-5 \end{cases}$$

$$\Rightarrow\begin{cases} x<16, & x\geq\frac{1}{2} \\ -3x<14, & -5\leq x<\frac{1}{2} \\ -x<4, & x<-5 \end{cases}$$

$$\Rightarrow\begin{cases} x<16, & x\geq\frac{1}{2} \\ x>-\frac{14}{3}, & -5\leq x<\frac{1}{2} \\ x>-4, & x<-5 \end{cases}$$

Case 1: When $x \geq \frac{1}{2}$, we have $x < 16$. Therefore, the inequality holds true if

$$\frac{1}{2} \leq x < 16 \text{ (same as } x \in \left[\frac{1}{2}, 16\right)\text{)}.$$

Case 2: When $-5 \leq x < \frac{1}{2}$, we have $x > -\frac{14}{3}$. Therefore, the inequality holds true if

$$-\frac{14}{3} < x < \frac{1}{2} \ \left(x \in \left(-\frac{14}{3}, \frac{1}{2}\right)\right).$$

Case 3: When $x < -5$, this is impossible.

Combining both Case 1 and 2, we got the inequality holds true if $-\frac{14}{3} < x < 16$.

Exercise: Solve $\left|x+2\right|+\left|x-2\right|-\left|x\right|\leq 4$. Ans: $x\in[-4,4]$

Theorem 3.3: Properties of Absolute Value Function

(i) $|a| \geq 0$, with equality if and only if $a = 0$.

(ii) $|a| = |-a|$.

(iii) $-|a| \leq a \leq |a|$.

(iv) $|a \cdot b| = |a| \cdot |b|$.

(v) $1/|a| = |1/a|$, if $a \neq 0$.

(vi) $|a|/|b| = |a/b|$, if $b \neq 0$.

(vii) $|a| \leq b$ if and only if $-b \leq a \leq b$.

(viii) (Triangle Inequality) $|a + b| \leq |a| + |b|$.

(ix) (Reverse Triangle Inequality) $||a| - |b|| \leq |a - b|$.

Proof of the triangle inequality.

Let a and b be real numbers.

$a+b \le \lvert a \rvert + \lvert b \rvert$	from (iii)
$\& -(a+b) = -a-b \le \lvert -a \rvert + \lvert -b \rvert = \lvert a \rvert + \lvert b \rvert$	from (ii)
$\therefore \lvert a+b \rvert \le \lvert a \rvert + \lvert b \rvert$	Definition for absolute value involving inequalities (case 1)

Self check

What is a field?

Let $(\mathbb{F}, +, \cdot)$ be a system, where $\mathbb{F}$ is a non-empty set equipped with two operations addition $(+)$ and multiplication $(\cdot)$. If each and every of the following laws are satisfied:

Closed for both operations (binary operations)

Associative law for both operations.

Identity law for both operations.

Inverse law for both operations.

Commutative law for both operations.

Distributive law.

Then $\mathbb{F}$ is a field.

Self check

What is an ordered field?

Suppose now $\mathbb{F}$ is a field, and in addition the system has two relations $(<)$ and $(=)$.

If $\mathbb{F}$ fulfils the following law:

Order Axioms

Then $\mathbb{F}$ is an ordered field. We can see that an ordered field is a system with an additional rule on a field.

Self check

What is a complete ordered field?

Suppose now $\mathbb{F}$ is an ordered field. In addition, there exists supremum for any bounded subset of $\mathbb{F}$. Then $\mathbb{F}$ is a complete ordered field.

Self check

How many complete ordered fields we have?

We can have plenty of them, but they are all *isomorphic*.

We name $\mathbb{R}$ (the set of real numbers) to be this unique complete ordered field.

Self check

We now may define the irrational numbers as well:

$$\mathbb{I} = \mathbb{Q}' := \mathbb{R} \setminus \mathbb{Q} \qquad \text{the set of } \textit{irrational numbers}$$

Summary of max/min, LB/UB, sup/inf

Let S be a subset of an ordered field.

	Does it always exist?	Unique (if it exist)?	An element of S?
Max/min of S	No	Yes	Yes
Upper/lower bound of S	No	No	Not necessary
Sup/inf of S	No, but Yes if ...*	Yes	Not necessary

Note: Yes if S is a subset of a complete ordered field and S is bounded above/below.

Back to our very first question…

What we want?



Define a number system that can measure any length.



We want not just positive, but also zero and negative numbers.



Define a number system that can fully fills the number line.

Definition of density

Let $A, B \neq \phi$ with $A \subseteq B$. Then A is said to be dense in B if

$$\forall b_1, b_2 \in B \text{ with } b_1 < b_2, \exists a \in A \ni b_1 < a < b_2.$$

Density of real numbers

We have showed that rational numbers are dense. What about the real numbers?

To prove that real numbers are dense, we need to introduce the following two important theorems.

Theorem 3.4. (Archimedean Property) $\forall x, y \in \mathbb{R}^+$ with $y > x, \exists n \in \mathbb{N} \ni y < nx$.

Corollary 3.5 (Archimedean Property) $\forall x \in \mathbb{R}^+$, $\exists n \in \mathbb{N} \ni x > \frac{1}{n}$.

Videos are available for the proofs.

Theorem 3.7. (Density of $\mathbb{Q}$ in $\mathbb{R}$) $\forall x, y \in \mathbb{R}$ with $x < y$, $\exists z \in \mathbb{Q} \ni x < z < y$.

Corollary 3.8. (Density of $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$) $\forall x, y \in \mathbb{R}$ with $x < y$, $\exists z \in \mathbb{R} \setminus \mathbb{Q} \ni x < z < y$.

The above two results show that $\mathbb{R}$ is dense, i.e.,

$$\forall x, y \in \mathbb{R} \text{ with } x < y, \exists z \in \mathbb{R} \ni x < z < y.$$

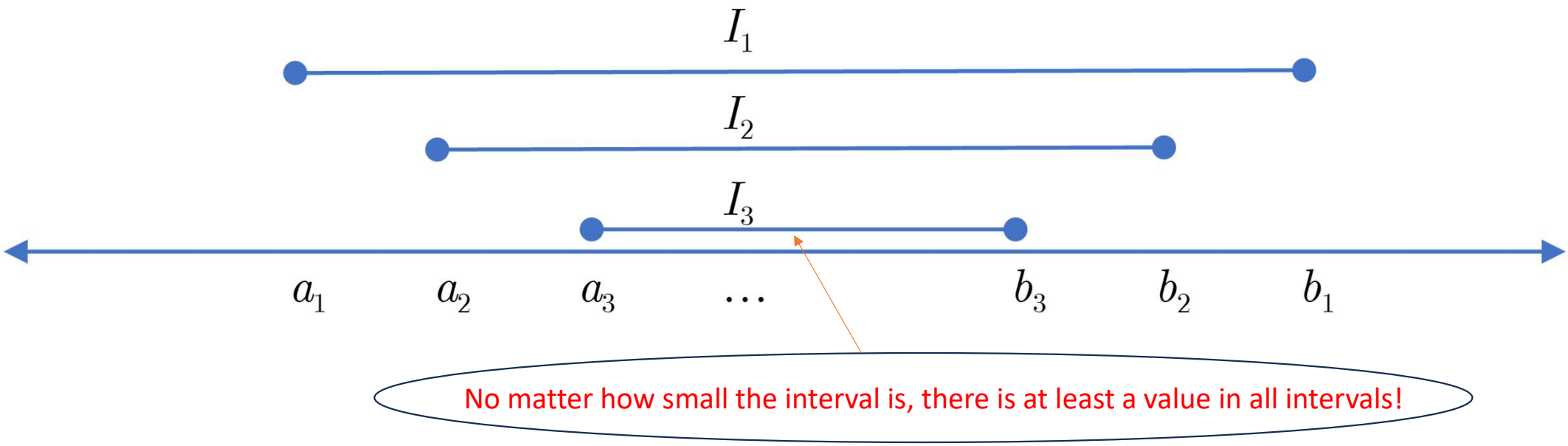
Videos are available for the proofs.

Nested Interval Property

Theorem 3.6: Nested Interval Property

Let $I_n = [a_n, b_n] = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}$ be a closed internal with $I_{n+1} \subset I_n$ for each $n \in \mathbb{N}$. Then there exist an $x \in I_n$ for every $n \in \mathbb{N}$.

Videos are available for the proofs.



We have got the final piece done! The set of real numbers is constructed. You are now free to use it without any doubt!

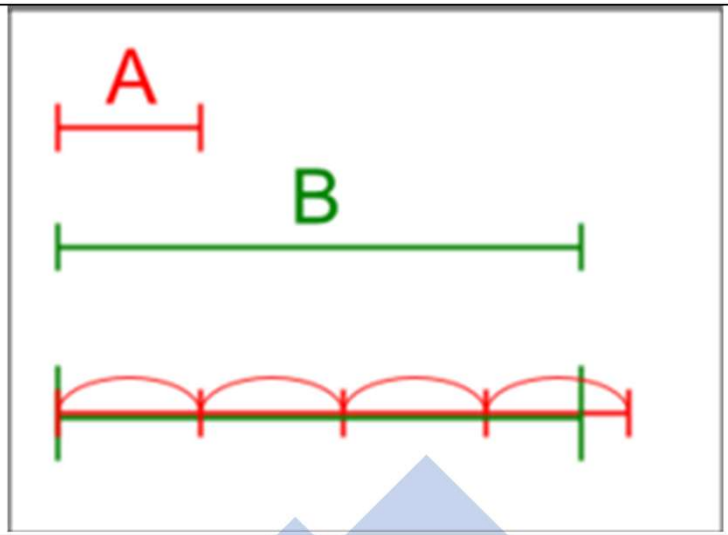
Archimedean property

1. State the Archimedean property.

2. Prove Archimedean property.

3. Where to apply Archimedean property? Or why we learn Archimedean property?

Answer to 3. To prove that $\mathbb{N}$ is unbounded; to avoid having exotic numbers in $\mathbb{R}$, such as surreal numbers and infinitesimal numbers; and can be applied in many proofs.

Further reading for your perusal Infinitesimal - Wikipedia Surreal number - Wikipedia Archimedean property - Wikipedia	

Density of real number

1. State the meaning of the Density of $\mathbb{R}$.

2. Prove that $\mathbb{R}$ is dense.

3. $\mathbb{R}$ is dense, so what this means?

Answer to 3. If $\mathbb{R}$ is dense, it means that there are infinitely many numbers on the real line, and it can be extended to infinity. This would be sufficient for length measurements. However, there are other measure in Mathematics (the branch is called the measure theory) read Measure (mathematics) - Wikipedia.

Nested Interval property (NIP)

1. State NIP.
2. Prove NIP.
3. Why we learn NIP?

Answer to 3. NIP can be applied in many ways; ancient Babylonian discovered a method for computing square roots using this property. In our context, we apply NIP to show that the real line is continuous. [*Nested intervals - Wikipedia*]

Recall

Let S be a nonempty subset of an ordered field $\mathbb{F}$.

Definition of Upper bound: Then $U \in \mathbb{F}$ is an upper bound of S if $U \geq s$ for all $s \in S$.

Definition of Lower bound: Then $L \in \mathbb{F}$ is a lower bound of S if $L \leq s$ for all $s \in S$.

$A = \left\{x \in \mathbb{Q} : -\sqrt{2} < x < \sqrt{2}\right\}$, meaning we can choose all the rational numbers in between $-\sqrt{2}$ and $\sqrt{2}$.

Example. Consider the $A = \left\{x \in \mathbb{Q} : x^2 < 2\right\} \subset \mathbb{Q}$. Then we see that

Upper bound	$\frac{3}{2}$, 2, etc
Lower bound	-2, -3, etc

Among the (infinitely many) upper bounds, can we find the smallest number among them? Does it even exist?

Exercise

Give an example of upper bound and lower bound, if it exists:

$$S = \{x \in \mathbb{Z}^+ : (x-1)(x+2) < 0\} \subset \mathbb{Q}\,.$$

Completeness Axiom

- Let S be a nonempty subset of a complete ordered field, if S is bounded above. Then Sup S is the LEAST upper bound of S. Sup S exists and it is unique.

- Let S be a nonempty subset of a complete ordered field, if S is bounded below. Then inf S is the GREATEST lower bound of S. Inf S exists and it is unique.

Theorem 4.1. Uniqueness of Supremum

Let S be a nonempty subset of $\mathbb{R}$ that is bounded above.

Then (by the complete axiom) there exists a supremum of S.

Moreover, the supremum is unique.

Videos are available for the proofs.

Similarly, let S be a nonempty subset of $\mathbb{R}$ that is bounded below. Then (by the complete axiom) there exists an infimum of S. Moreover, the infimum is unique.

Example. Suppose set $A := \left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\} \subset \mathbb{R}$. Show that

(a) $\sup A = 1$ and (b) $\inf A = 0$ (exercise)

Solution.

(a) We prove that • A is nonempty (this is obvious)

- 1 is an upper bound of A; and
- 1 is lesser than any other upper bound of A.

Proof. It is obvious that 1 is an upper bound of A because $1 > 1 - \frac{1}{n}$ $\forall n \in \mathbb{N}$.

Recap: Archimedean property

Theorem 3.4. For all positive real numbers a and b such that $a > b$, there exists a natural number n such that $a < nb$.

Writing in the notation of mathematics: $\forall a, b \in \mathbb{R}^+,\ a > b,\ \exists n \in \mathbb{N} \ni a < nb$.

Corollary 3.5. For all positive real numbers r, there exists a natural number n such that $r > 1/n$.

Writing in the notation of mathematics: $\forall r \in \mathbb{R}^+,\ \exists n \in \mathbb{N} \ni r > \frac{1}{n}$.

Next, we prove that 1 is lesser than any other upper bound of A (by contradiction). Suppose not, we let $m < 1$ be the supremum. Write $m = 1 - \varepsilon$, for some $\varepsilon > 0$. Now since

$$m \geq 1 - \tfrac{1}{n} \; \forall n \in \mathbb{N},$$

(because m is an upper bound), it implies that

$$1 - \varepsilon \geq 1 - \tfrac{1}{n} \Rightarrow \varepsilon \leq \tfrac{1}{n} \; \forall n \in \mathbb{N}.$$

This is a contradiction because by the Archimedean property, we know that there exists natural number n such that $\frac{1}{n} < \varepsilon$.

Example. Let $B = \{n-1 : n \in \mathbb{N}\}$. Then $\inf B = 0$.

Proof. Note that $0 \leq n-1 \ \ \forall n \in \mathbb{N}$. So, 0 is a lower bound. Now suppose $\inf B = m$, and we further assume that $m > 0$.

Now we have $m \leq n-1 \ \forall n \in \mathbb{N}$. Thus, $0 < m \leq n-1 \ \forall n \in \mathbb{N}$. Which means, $n > 1$ $\forall n \in \mathbb{N}$. This is a contradiction because we can take $n = 1$.

Theorem 4.2. Suppose A is a nonempty set.

(i) Let $\alpha \in \mathbb{R}$ be an upper bound for a set $A \subseteq \mathbb{R}$. Then

$$\alpha = \sup A \Leftrightarrow \forall \varepsilon > 0 \ \exists a \in A \ni \alpha - \varepsilon < a .$$

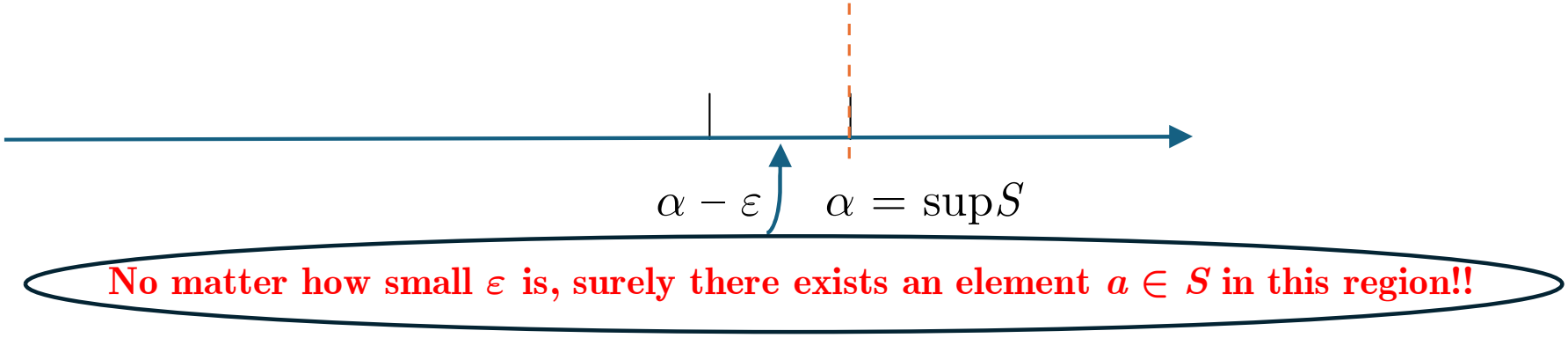
(ii) Let $\beta \in \mathbb{R}$ be a lower bound for a set $B \subseteq \mathbb{R}$. Then

$$\beta = \inf A \Leftrightarrow \forall \varepsilon > 0 \ \exists b \in B \ni \beta + \varepsilon > b .$$

Remark: This is method way to show a value is the sup/inf of a set.

Videos are available for the proofs.

Theorem 4.2(i) Let $\phi \neq S \subseteq \mathbb{R}$ be bounded above (so that $\sup S$ exists). Let α be an upper bound of S. Then $\sup S = \alpha$ if and only if there exists an element a in S such that $\alpha - \varepsilon < a$ for all $\varepsilon \in \mathbb{R}^+$.



Question 1: What if sup $S \in S$

Question: $S = \{x < 2\text{: } x \text{ is integer}\}$. Can you find an element $a \in S$ such that $\alpha - \varepsilon < a$ for all positive real number ε?

Answer: Then sup $S = 1$. Take $a = 1$, then $1 - \varepsilon < 1$ for all $\varepsilon > 0$.

Meaning to say:

If supremum is an element of the set S, then one can take $a = \sup S$, and

$$a - \varepsilon < a \text{ is true for all } \varepsilon \in \mathbb{R}^+.$$

Question 2: How to use Theorem 4.2?

Example. Suppose set $A := \left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\} \subset \mathbb{R}$. Show that

(a) $\sup A = 1$ and (b) $\inf A = 0$ (exercise)

Let $a \in A$. Suppose there is absolutely no a such that $1 - \varepsilon < a$ for some $\varepsilon > 0$. Then

$$1 - \varepsilon \geq a = 1 - \tfrac{1}{n} \Rightarrow \varepsilon \leq \tfrac{1}{n} \text{ for all } n \in \mathbb{N}.$$

But this is contradicting to the Archimedean property (Corollary 3.5), thus, we have proven that there exists at least an element a in A such that $1 - \varepsilon < a$ for all $\varepsilon > 0$. Applying Theorem 4.2(i), we conclude that $\sup A = 1$.

Definition 4.4. Maximum and minimum of a set

Let $\phi \neq A \subseteq \mathbb{R}$. Then if there exists an element $M \in A$ such that

$$a \leq M \ \forall a \in A,$$

A is said to have a maximum denoted by $\max A \ (= M)$.

Similarly, we can define the minimum of a set. (Exercise)

Example: Suppose set $A := \left\{1 + \frac{1}{n} : n \in \mathbb{N}\right\}$.

(a) Is the minimum of A exist? Justify your answer.

(b) Is the min $A = \inf A$? Justify your answer.

Solution.

(a) The minimum of A does not exist because there is no element in A such that the element is smaller than all elements in A. Note that $1 \notin A$.

(b) The minimum is not equals to 1 (inf of A) simply because it does not exist. Note that if minimum of a set exists, then surely infimum is equal to this value.

Definition. Let $a, b \in \mathbb{R}$ such that $a \neq b$. Then

$$\max\{a,\, b\} := \begin{cases} a, & \text{if } a > b \\ b, & \text{if } b > a \end{cases}.$$

For example, $\max\{1,\, 2\} = 2$, $\max\{0,\, 0\} = 0$, $\max\{-1, -2\} = -1$ and $\max\{\pi, e\} = \pi$.

Remark: $\min\{a,\, b\}$ is defined similarly; it is the smaller one of a and b.

Example 4.5. Prove that $\sqrt{2} \in \mathbb{R}$.

Videos are available for the proofs.

We have proven that square root of 2 is not rational,
but how we show that it is irrational?
Meaning to say, how do we show that square root of 2
is indeed a real number and not something else?